Grass

Solutions to Baby Rudin / PMA

January 2024-Today

Contents

1 The Real and Complex Number Systems **2**

2 Basic Topology **6**

3 Numerical Sequences and Series **17**

4 Bibliography **19**

Chapter 1

The Real and Complex Number Systems

Exercise 1.1. If r is rational ($r \neq 0$) and x is irrational, prove that $r + x$ and rx are irrational.

Proof. If they were rational, $(r + x) - r$ and $rx \cdot 1/r$ would also be rational, a contradiction.

Exercise 1.3. Prove Proposition 1.15.

Proof. (a) The axioms (M) give

$$\begin{aligned}
(1/x)(xy) &\overset{\text{M5}}{=} (1/x)(xz) \\
[(1/x)x]y &\overset{\text{M3}}{=} [(1/x)x]z \\
[x(1/x)]y &\overset{\text{M2}}{=} [x(1/x)]z \\
1 \cdot y &\overset{\text{M5}}{=} 1 \cdot z \\
y &\overset{\text{M4}}{=} z
\end{aligned}$$

(b) Fix $z = 1$ in (a).
(c) Take $z = 1/x$ in (a).
(d) Apply (c) to $(1/x)x = 1$.

Exercise 1.5. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf(A) = -\sup(-A).$$

Proof. Let $i := \inf(A)$ and $s = \sup(-A)$. Notice $-i$ is a upper bound of $-A$, and $-s$ is a lower bound of A. As such, $-i \geq s$ and $i \geq -s$; the equality $i = -s$ holds.

Exercise 1.6. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are irrational.

(c) If x is real, define $\mathrm{B}(x)$ to be the set of all numbers b^t, where t is rational and $t \leq x$. Prove that

$$b^r = \sup \mathrm{B}(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup \mathrm{B}(x)$$

for every real x.

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y.

Proof. (a) Notice $b^{mq} = b^{pn}$. By taking (nq)th roots, we have

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

(b) By the Corollary to Theorem 1.21,

$$b^{\frac{m}{n}+\frac{p}{q}} = b^{\frac{mq+pn}{nq}} = (b^{mq}b^{pn})^{\frac{1}{nq}} = b^{\frac{m}{n}}b^{\frac{p}{q}}.$$

(c) Let $m/n > p/q$. Then $b^{m/n} > b^{p/q}$, lest $b^{mq} \leq b^{pn}$ (a contradiction). The converse hence holds. So, $b^r = \sup \mathrm{B}(r)$.

(d) For rational t, note $b^{x+t}b^{-t}$ bounds $\mathrm{B}(x)$ from above. That is, $b^{x+t} \geq b^x b^t$. Equality holds since $b^x b^y \geq b^{x+y}$ is clear. Let $s := \sup\{b^{x+t} \mid t \leq y\}$. Then, $s(b^x)^{-1}$ bounds $\mathrm{B}(y)$ from above. As such, $s \geq b^x b^y$. So $b^x b^y \leq s \leq b^{x+y}$.

Exercise 1.8. Prove that no order can be defined in the complex field that turns it into an ordered field. *Hint:* -1 is a square.

Proof. Suppose, for contradiction, that $(\mathbb{C}, <)$ is an ordered field. Wlog, let $1 > -1$. Then $i^2 = (-i)^2 = -1 > 0$ so $-1 > 1$, a contradiction.

Exercise 1.10. Suppose $z = a + bi$, $w = u + iv$, and

$$a = \left(\frac{|w| + u}{2}\right)^{1/2}, \qquad b = \left(\frac{|w| - u}{2}\right)^{1/2}.$$

Proof. If $v \geq 0$, then $z^2 = u + |v|i = w$. Similarly when $v \leq 0$, we have $(\bar{z})^2 = u - |v|i = w$. We conclude that every nonzero complex number has two complex square roots.

Exercise 1.12. If $z_1, \ldots, z_n$ are complex, prove that

$$|z_1 + z_2 + \cdots + z_n| \leq |z_1| + |z_2| + \cdots + |z_n|.$$

Proof. The result is clear by using induction on the triangle inequality (Theorem 1.33(e)).

Exercise 1.13. If x, y are complex, prove that

$$\big||x| - |y|\big| \leq |x - y|.$$

Proof. Clearly, square roots of reals preserve order. It suffices to notice

$$\begin{aligned} |x - y|^2 &= (x - y)(\bar{x} - \bar{y}) \\ &= x\bar{x} - \mathrm{Re}(x\bar{y}) + y\bar{y} \\ &\geq |x|^2 - 2|x\bar{y}| + |y|^2 \\ &= |x|^2 - 2|x||y| + |y|^2 \\ &= ||x| - |y||^2. \end{aligned}$$

Note 1.14. The *real* case is marginally simpler:

$$\big||x| - |y|\big|^2 = x^2 - 2|x||y| + y^2 \leq x^2 - 2xy + y^2 = |x - y|^2.$$

Exercise 1.15. Under what conditions does equality hold in the Schwarz inequality?

Proof. Consider the real case. Suppose that there exists $x \in \mathbb{R}$, so for all i, so $b_i = xa_i$. Then, equality is clear as

$$\left|\sum_{i=1}^{n} a_i b_i\right|^2 = x^2 \left|\sum_{i=1}^{n} a_i^2\right|^2 = \sum_{i=1}^{n} |a_i|^2 \sum_{i=1}^{n} |xa_i|^2.$$

Exercise 1.17. Prove that

$$|\mathbf{x}+\mathbf{y}|^2 + |\mathbf{x}-\mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2$$

if $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^k$. Interpret this geometrically, as a statement about parallelograms.

Proof. Notice that

$$\begin{aligned} |\mathbf{x}+\mathbf{y}|^2 &= (\mathbf{x}+\mathbf{y})\cdot(\mathbf{x}-\mathbf{y}) \\ &= |\mathbf{x}|^2 + 2\mathbf{x}\cdot\mathbf{y} + |\mathbf{y}|^2, \end{aligned} \qquad \text{and} \qquad \begin{aligned} |\mathbf{x}-\mathbf{y}|^2 &= (\mathbf{x}-\mathbf{y})\cdot(\mathbf{x}-\mathbf{y}) \\ &= |\mathbf{x}|^2 - 2\mathbf{x}\cdot\mathbf{y} + |\mathbf{y}|^2. \end{aligned}$$

Therefore, $|\mathbf{x}+\mathbf{y}|^2 + |\mathbf{x}-\mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2$. Let A be the area of the two squares having length equal to a parallelogram's respective diagonals. Also let B be the area of the two squares having side lengths equal to the parallelogram's longer and shorter sides. Then, our equality implies $A = 2B$.

Proof. Alternatively, this can be proved as a corollary of Pythagoras' Theorem. We know

$$|\mathbf{x}+\mathbf{y}|^2 = |x|^2 + |y|^2 \qquad \text{and} \qquad |\mathbf{x}-\mathbf{y}|^2 = |x|^2 + |y|^2.$$

Taking the sum,

$$|\mathbf{x}+\mathbf{y}|^2 + |\mathbf{x}-\mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2.$$

Chapter 2

Basic Topology

Theorem 2.30. Suppose $Y \subseteq X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X.

Proof. Let $p \in E$.
Assume E is open relative to Y. Then, there is $r > 0$ so $N_r(p) \cap Y \subseteq E$. Notice, $N_r(p) \cap (Y - E) = \varnothing$. So we define $G := \bigcup_{p \in E} N_{r_p}(p)$. Clearly, this superset of E is open relative to X.
Conversely, suppose $E = Y \cap G$ for some open subset G of X. Then, as $p \in G$, there is $r > 0$ with $N_r(p) \subseteq G$. As such, $N_r(p) \cap Y \subseteq E$. Hence, E is open relative to Y.

Example 2.31. Let $X = \mathbb{R}^2$. In each case, G is the combined region enclosed by the dotted lines, and Y consists of the shaded area and all points of E.

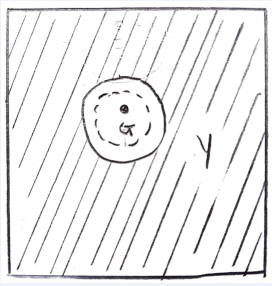

Figure 2.1: E contains only the point in black.

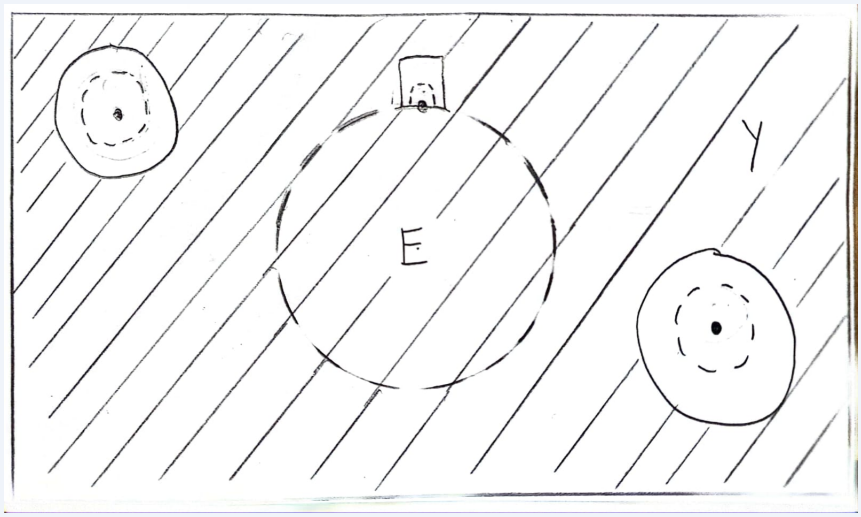

Figure 2.2: E contains the central dotted circle and the points in black.

Theorem 2.33. Suppose $K \subseteq Y \subseteq X$. Then, K is compact relative to x if and only if K is compact relative to Y.

Proof. When K is compact relative to X, let $\{F_\alpha\}$ be an open cover in Y. By theorem 2.30, there is an open cover $\{G_\alpha\}$ in X containing a finite subcover $\{G_{\alpha_n}\}$, for which $Y \cap G_\alpha$ is always F_α. So, $\{F_{\alpha_n}\}$ is a finite subcover of $\{F_\alpha\}$.
Conversely, assume K is compact relative to Y and let $\{G_\alpha\}$ be an open cover in X. Then, $\{Y \cap G_\alpha\}$ is an open cover in Y. As such, it contains a finite subcover $\{Y \cap G_{\alpha_n}\}$ in $Y \subseteq X$.

Theorem 2.34. Compact subsets of metric spaces are closed.

Proof. Let $x \in E^{\complement}$ be a limit point of E. Now, define the open cover

$$\mathcal{C} := \{N_{d(x,p)/2}(p) \,|\, p \in E\}.$$

Given any finite subset $\{N_{d(x,p)/2}(p_i) \,|\, 1 \leq i \leq n\}$, fix $m > 0$ as the minimum distance x is from all such p_i. Then, there is a point v in E such that $d(x, v) < m/2$. No finite subcover of $\mathcal{C}$ exists; hence E is not compact.

Theorem 2.35. Any closed subset of a compact set is compact.

Proof. Let E be a closed subset of the compact set F, and $\{G_\alpha\}$ be an open cover of E. Then, F has a finite subcover $\{G_{\alpha_n} \cup E^{\complement}\}$. So, $\{G_{\alpha_n}\}$ must be a finite subcover of E.

Theorem. Finite unions of compact sets are compact.

Proof. Let $\{K_i\}_{i=1}^n$ contain only compact sets, and $\{G_\alpha\}$ be an open cover of $\mathcal{K} := \bigcup_i K_i$. Each K_i has a finite subcover $\{G_{\alpha_{ij}} \mid 1 \leq j \leq m_i\}$. So,

$$\{G_{\alpha_{ij}} \mid 1 \leq i \leq n \text{ and } 1 \leq j \leq m_i\}$$

is a finite subcover of $\mathcal{K}$.

Note. Infinite unions of compact sets aren't necessarily compact.

Example. Consider the metric space $\mathbb{R}$. Clearly, each singleton set $\{x\}$ is compact. But since the set

$$(0,1] = \bigcup \{\{x\} \mid x \in (0,1]\}$$

does not contain the limit point 0, it is not compact.

Theorem 2.36. If $\{K_\alpha\}$ is a collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap_\alpha K_\alpha$ is nonempty.

Proof. Assume, for contradiction, that finite intersections in $\{K_\alpha\}$ are nonempty, but $\bigcap_\alpha K_\alpha = \varnothing$. Since $\{K_\alpha\}$ can't be empty, there is some K_β. It has the open cover $\left\{K_\alpha^{\complement}\right\}$.

Consider any finite subset $\left\{K_i^{\complement}\right\}$. Then, it is disjoint from the nonempty set $K_\beta \cap \bigcap_i K_i$. As such, K_β is not compact. A contradiction.

Corollary. Let $\{K_\alpha\}$ be a set of closed sets, at least one of which is compact, such that $\{K_\alpha\}$ has the finite intersection property. Then, $\bigcap_\alpha K_\alpha$ is nonempty.

Proof. Wlog, K_0 is compact. Then, $\{K_0 \cap K_\alpha\}$ is a set of nonempty compact sets. Hence, the preceeding theorem says $\bigcap_\alpha K_\alpha$ is nonempty.

Definition. We say that a set (E,d) is *globally closed* iff for all supersets (X,d') of (E,d) (such that there is an isometric embedding of (E,d) in (X,d')), we have that E is closed relative to X.

Claim. If a set (E,d) is globally closed, it is compact.

Here is a counterexample.

Example. Consider the non-compact metric space $\mathbb{N}$ under the Euclidean norm. Suppose, for contradiction, that there exists some superset X in which $\mathbb{N}$ is not closed (such that there is an isometric embedding of $\mathbb{N}$ in X).

Let p be such a limit point. So, there exists $n_2 < n_3$, such that $d(n_2, x)$ and $d(n_3, x)$

are less than $1/2$. Consequently,

$$d(n_2, x) + d(x, n_3) < 1 \leq d(n_2, n_3).$$

A contradiction.

Thus, compactness is quite a strong condition; it is stronger than even global closure!

Theorem 2.37. If E is an infinite subset of a compact set K, then E has a limit point in K.

Proof. Suppose, for contradiction, that there exists a subset E of some compact set K, without a limit point in K. As such, E is closed. Hence, it is compact. Since E contains no limit points, for each $p \in E$ there is a minimum distance m_p, such that $m_p \leq d(p, q)$ for all $q \in E$ (that isn't p). But now the open cover $\{N_{m_p/2}(p) \mid p \in E\}$ has no finite subcover. A contradiction.

Theorem 2.38. If $\{I_n\}_{n=1}^{\infty}$ is a sequence of closed intervals in $\mathbb{R}$, such that $I_{n+1} \subseteq I_n$, then $\bigcap_n I_n$ is nonempty.

Proof. Let $I_n = [a_n, b_n]$. Then, $a_n \leq \sup\{a_n\}_{n=1}^{\infty} \leq b_n$ for all n.

Theorem 2.39. Let k be a positive integer. If $\{I_n\}$ is a sequence of k-cells such that $I_{n+1} \subseteq I_n$, then $\bigcap_n I_n$ is nonempty.

Proof. Similarly, let I_n be the k-cell of elements $(x_1, x_2, \ldots, x_k)$, such that $a_{ni} \leq x_i \leq b_{ni}$. Also let $s_i := \sup\{a_{ni}\}_{n=1}^{\infty}$. Notice that $a_{ni} \leq s_i \leq b_{ni}$ for all n and i. That is, $(s_1, s_2, \ldots, s_k) \in \bigcap_n I_n$.

Theorem 2.40. Every k-cell is compact.

Proof. Let $\{G_\alpha\}$ be an open cover of a k-cell I.
Notice that for each $\mathbf{x} \in I$, there exists $m(\mathbf{x}) > 0$, such that $N_{m(\mathbf{x})}(\mathbf{x})$ is contained in some G_α and

$$m(\mathbf{x}) > \frac{1}{2} \sup\{r \mid N_r(\mathbf{x}) \subseteq G_\alpha \text{ for some } \alpha\}.$$

Suppose, for contradiction, that $\ell := \inf\{m(\mathbf{x}) \mid \mathbf{x} \in E\} = 0$. By AC, we can choose a sequence $\{\mathbf{x}_n\}_{n=1}^{\infty}$ for which $\mathbf{x}_n := (x_{1n}, x_{2n}, \ldots, x_{kn})$ and $m(\mathbf{x}_n) < 1/n$. By iteratively applying Bolzano-Weierstrass, there exists a subsequence $\{\mathbf{x}_{n_k}\}_{k=1}^{\infty}$ such that $s_i := \lim_{k\to\infty} x_{n_k}$ always exists. Let $\mathbf{s} \in I$ be the point whose ith

coordinate is s_i. So, for some $n \in \mathbb{N}$, we have $2m(\mathbf{x}_n) < m(\mathbf{s})$, satisfying

$$|s_i - x_{in}| < \frac{1}{\sqrt{k}}\big(m(\mathbf{s}) - 2m(\mathbf{x}_n)\big).$$

Hence,

$$\sum_{i=1}^{k}(s_i - x_{in})^2 < \sum_{i=1}^{k}\frac{1}{k}\big(m(\mathbf{s}) - 2m(\mathbf{x})\big)^2 = \big(m(\mathbf{s}) - 2m(\mathbf{x}_n)\big)^2.$$

That is, $|\mathbf{s} - \mathbf{x}_n| < m(\mathbf{s}) - 2m(\mathbf{x}_n)$. Now for each $|\mathbf{x}_n - \mathbf{q}| < 2m(\mathbf{x}_n)$,

$$|\mathbf{s} - \mathbf{q}| \leq |\mathbf{s} - \mathbf{x}_n| + |\mathbf{x}_n - \mathbf{q}| < m(\mathbf{s}) - 2m(\mathbf{x}_n) + 2m(\mathbf{x}_n) = m(\mathbf{s}).$$

As such, $N_{2m(\mathbf{x}_n)}(\mathbf{x}_n)$ is contained in the same G_α as $N_{m(\mathbf{s})}(\mathbf{s})$. But by definition

$$2m(\mathbf{x}_n) > \sup\{r \mid N_r(\mathbf{x}) \subseteq G_\alpha \text{ for some } \alpha\},$$

a contradiction.
Consequently, $\ell > 0$ so there exists a finite number of neighbourhoods (of radius ℓ), each contained in some G_{α_n}, that covers I.

Oh, actually I probably don't need to use Bolzano-Weierstrass. It's pretty clear that there must be coordinate-wise convergence, in order for $\{\mathbf{x}_n\}_{n=1}^{\infty}$ to converge.

Theorem 2.41. If a set in $\mathbb{R}^k$ has one of the following properties, then it has the other two:

(a) E is closed and bounded.
(b) E is compact.
(c) Every infinite subset of E has a limit point in E.

Proof.

i. Assume (a) is true. Then, let

$$a_i := \inf\{x_i \mid (x_1, x_2, \ldots, x_k) \in \mathbb{R}^k\} \quad \text{and} \quad b_i := \sup\{x_i \mid (x_1, x_2, \ldots, x_k) \in \mathbb{R}^k\}.$$

Now, E is a closed subset of the k-cell of points $(x_1, x_2, \ldots, x_k)$, such that $a_i \leq x_i \leq b_i$. Since k-cells are compact, so is E. That is, (b) holds.

ii. Now suppose (b) is valid. Thus, (c) follows immediately from theorem 2.37.

iii. Finally, we presume (c) to be true. If E is unbounded, then (using AC) we can construct a sequence $\{\mathbf{x}_n\}_{n=1}^{\infty}$ of points, such that $|\mathbf{x}_{n+1}| > |\mathbf{x}_n| + 1$ for all n. Hence, it is clear that $\{\mathbf{x}_n \mid n \in \mathbb{N}\}$ contains no limit points. A contradiction. Similarly, when $\mathbf{p} \notin E$ is a limit point of E, there is a sequence $\{\mathbf{y}_n\}_{n=1}^{\infty}$, with $|\mathbf{p} - \mathbf{y}_n| < 1/n$. Therefore, $\{\mathbf{y}_n \mid n \in \mathbb{N}\}$ also has no limit points in E. Again, this is impossible; (a) must hold.

Question. When a set is unbounded, must it be non-compact?

Claim. (Revised) If a set (E, d) is both globally closed and bounded, it is compact.

Theorem 2.42. Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

Proof. Let K be a bounded infinite subset of $\mathbb{R}^k$. Then, its closure $\overline{K}$ is compact. Hence, there exists a limit point of $\overline{K}$. This is also a limit point of K.

Exercise. (JohnDS's Exercise) Let X be a metric space. Prove the following are equivalent.

(1) $E \subseteq X$ is dense.
(2) For every $\varepsilon > 0$ and all $x \in X$, there is an $p \in E$ such that $d(x, p) \leq \varepsilon$.
(3) The closure of E is X.
(4) For every $x \in X$, there is a sequence $\{p_n\}_{n=1}^{\infty}$ in E, such that $x = \lim_{n\to\infty} p_n$.

Proof. Clearly, (3) implies (1).
Now assume (1) is true. Pick any $x \in X$ and $\varepsilon > 0$. If $x \in E$, simply let $p = x$. Otherwise, x is a limit point of E. Thus, such p exists by definition. As such, (2) holds.
Suppose (2) is true. By AC, there is a sequence $\{p_n\}_{n=1}^{\infty}$ with $d(x, p_n) < 1/n$. That is, $x = \lim_{n\to\infty} p_n$. So, (4) holds.
Presume (4) is satisfied. When $p_n = x$ for some n, we know $x \in E$. Otherwise, p_n is never x, which means x is a limit point of E. Consequently, (3) holds true.

Exercise 2.5. Construct a bounded set of real numbers with exactly three limit points.

Proof. Let $E := \{1/n, 2/n, 3/n \,|\, n \in \mathbb{Z}^+\}$. By the Archimedean property, $\{1, 2, 3\} \subseteq E'$. In fact, equality is clear: the neighbourhood of any i/n (for $n \geq 2$) with radius

$$\min\left\{\frac{i}{n-1} - \frac{i}{n}, \frac{i}{n} - \frac{i}{n+1}\right\}$$

is disjoint from E.

Exercise 2.6. Let E' be the set of all limit points of a set E. Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall that $\overline{E} = E \cup E'$.) Do E and E' always have the same limit points?

Proof. Let p be a limit point of E'. So, for any $r > 0$, there is $q \in N_r(p) \cap E'$. Hence, letting $m := \min\{d(p, q), r - d(p, q)\}$, there is also $s \in N_m(q) \cap E$. As such,

p is a limit point of E, since

$$0 < d(p,s) \leq d(p,q) + d(q,s) < r.$$

Similarly, any limit point of $\overline{E}$ is a limit point of E. The converse is clear from $E \subseteq \overline{E}$.
However, limit points of E are not necessarily limit points of E'. For instance, let $E := \{1/n \mid n \in \mathbb{Z}^+\} \subseteq \mathbb{R}$. Then, $E' = \{1\}$ whilst $E'' = \varnothing$.

Exercise 2.8. Is every point of every open set $E \subseteq \mathbb{R}^2$ a limit point of E? Answer the same question for closed sets in $\mathbb{R}^2$.

Proof. Yes, every point p of an open set $E \subseteq \mathbb{R}^2$ is a limit point of E. There is $m > 0$ with $N_m(p) \subseteq E$. Thus, for any $r > 0$, the set $N_r(p) \cap E$ must be nonempty, for it contains the nonempty set $N_r(p) \cap N_m(p)$.
But this does not extend to closed sets in $\mathbb{R}^2$. Consider the closed set $\{1\}$; it has no limit points.

Exercise 2.10. Let X be an infinite set. For $p \in X$ and $q \in X$, define

$$d(p,q) = \begin{cases} 1 & (\text{if } p \neq q) \\ 0 & (\text{if } p = q). \end{cases}$$

Prove that this is a metric. Which subsets of the resulting metric space are open? Which are closed? Which are compact?

Proof. Notice that (a) and (b) holds immediately, and (c) is clear as $d(p,r)+d(r,q)$ is either 1 or 2. Therefore, d is a metric on X.
Let $E \subseteq X$ and $p \in E$. Observe that $N_{1/2}(p) = \{p\} \subseteq E$. All subsets of X must be open, and, with exception of $\varnothing$, be not closed.
Lastly, E is compact iff E is finite. If E is compact but infinite, then the open cover of singleton subsets of E (i.e. $\{\{p\} \mid p \in E\}$) has no finite subcover. The converse is clear as this open cover is already finite.

Exercise 2.11. For $x \in \mathbb{R}^1$ and $y \in \mathbb{R}^1$, define

$$\begin{aligned} d_1(x,y) &= (x-y)^2, \\ d_2(x,y) &= \sqrt{|x-y|}, \\ d_3(x,y) &= |x^2 - y^2|, \\ d_4(x,y) &= |x - 2y|, \\ d_5(x,y) &= \frac{|x-y|}{1+|x-y|}. \end{aligned}$$

Determine, for each of these, whether it is a metric or not.

Proof.

1. Since $(1-0)^2 = 1$ is larger than $(1-0.5)^2 + (0.5-0)^2 = 0.5$, we have that d_1 is not a metric.
2. Parts (a) and (b) are immediate. Moreover, (c) also holds, because for any real numbers x, y and z:

$$\begin{aligned} |x-y| &\leq |x-z| + 2\sqrt{|x-z||z-y|} + |z-y|, \\ \sqrt{|x-y|} &\leq \sqrt{|x-z|} + \sqrt{|z-y|}. \end{aligned}$$

   That is, d_2 is a metric.
3. Part (a) doesn't hold as $|1^2 - (-1)^2| = 0$. Hence, d_3 is not a metric.
4. Part (a) doesn't hold as $|1 - 2(0.5)| = 0$. Therefore, d_4 is not a metric.
5. Again, parts (a) and (b) are clear. Notice

$$|x-y| \leq |x-z| + |z-y| + 2|x-z||z-y| + |x-y||x-z||z-y|.$$

   Hence, adding $|x-y||x-z| + |x-y||z-y| + |x-y||x-z||z-y|$ to both sides, then factoring,

$$\begin{aligned} |x-y|(1+|x-z|)(1+|z-y|) \leq (&|x-z| + 2|x-z||z-y| + |z-y|) \\ &(1+|x-y|). \end{aligned}$$

   Therefore,

$$\frac{|x-y|}{1+|x-y|} \leq \frac{|x-z|}{1+|x-z|} + \frac{|z-y|}{1+|z-y|}.$$

   Accordingly, d_5 is a metric.

Exercise 2.12. Let $K \subseteq \mathbb{R}$ consist of 0 and the numbers $1/n$, for $n = 1, 2, 3, \ldots$. Prove that K is compact directly from the definition (without using the Heine-Borel Theorem).

Proof. Let $\{G_\alpha\}$ be an open cover of K. Then, $N_{1/n}(0) \subseteq G_{\alpha_{n+1}}$, for some $n \in \mathbb{N}$ and α_{n+1}. Furthermore, given any $1 \leq i \leq n$, there exists α_i for which $1/i \in G_{\alpha_i}$. Hence, $\{G_{\alpha_i} \mid 1 \leq i \leq n+1\}$ is a finite subcover.

Exercise 2.16. Regard $\mathbb{Q}$ as the set of all rational numbers, as a metric space, with $d(p,q) = |p-q|$. Let E be the set of all $p \in \mathbb{Q}$ such that $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$, but that E is not compact. Is E open in $\mathbb{Q}$?

Proof. Notice $|p-q|$ is always less than $\sqrt{3} - \sqrt{2}$. Now consider any limit point l of E. Then $2 \leq l^2 \leq 3$, lest $N_{|l-\sqrt{2}|}(l)$ or $N_{|l-\sqrt{3}|}(l)$ is disjoint from E. Since there

is no rational square root of 2 or 3, we have $l \in E$.
We have that E is not closed relative to $\mathbb{R}$, since the limit point $\sqrt{2} \notin E$. Thus, E is not compact.
Finally, E is indeed open in $\mathbb{Q}$, because for all $p \in E$, either $N_{|p-\sqrt{2}|}(p)$ or $N_{|p-\sqrt{3}|}(p)$ is contained in E.

Exercise 2.17. Let E be the set of all $x \in [0,1]$ whose decimal expansion contains only the digits 4 and 7. Is E countable? Is E dense in $[0,1]$? Is E compact? Is E perfect?

Proof. Let $a = 0.4777\ldots$ and $b = 0.7444\ldots$
An $a \leq x \leq b$ cannot be obtained by changing only the first digit, since

$$0.777\cdots > b > a \quad \text{and} \quad 0.444\cdots < a < b.$$

It also can't be obtained by changing the first digit and some others, as

$$0.777\ldots 74 \geq b > a \quad \text{and} \quad 0.444\ldots 47 \leq a < b.$$

(The digits, after what has been written out explicitly, are omitted.)
Nor does keeping the first digit the same:

$$0.4777\ldots 74 < a < b \quad \text{and} \quad 0.7444\ldots 47 > b > a$$

Hence no such x exists; E is not dense.
Let the decimal expansion of the limit point L contain some other digit not 4 or 7, first in the jth digit. Then, there exists $c \in E$ such that $|c - L| < 10^{-j}$. But now

$$1 > 10^j |c - L| \geq 1.$$

A contradiction. As such, E must be closed, and hence compact.
Furthermore, E is perfect, thus uncountable[a]. For any $y \in E$, define $y^{(i)}$ to be the ith digit of y, and

$$y^{[i]} := \begin{cases} 4 & \text{if } y^{(i)} = 7, \\ 7 & \text{if } y^{(i)} = 4. \end{cases}$$

So, let z_n be the number whose ith digit is $y^{(i)}$ if $1 \leq i \leq n$, and $y^{[i]}$ otherwise. Then, for each n we have

$$|y - z_n| < \frac{4}{10^{n+1}} < \frac{1}{10^n}.$$

[a] Alternatively, we can use diagonalization, or the fact that ${}^{\mathbb{N}}\{4,7\} \approx \mathscr{P}(\mathbb{N}) \succ \mathbb{N}$.

Exercise 2.22. A metric space is called *separable* if it contains a countable dense subset. Show that $\mathbb{R}^k$ is separable.

Proof. Let $\mathbf{x} = (x_1, x_2, \ldots, x_k) \in \mathbb{R}^k$. Recall $\mathbb{Q}$ is dense in $\mathbb{R}$. So, for each i, there exists $|x_i - p_i| < \frac{\varepsilon}{\sqrt{k}}$. Therefore,

$$|\mathbf{x} - \mathbf{p}| = \sqrt{\sum_{i=1}^{k} (x_i - p_i)^2} < \sqrt{\sum_{i=1}^{k} \frac{\varepsilon^2}{k}} = \varepsilon.$$

Since $\mathbb{R}^k$ contains the countable dense subset $\mathbb{Q}^k$, it is separable.

Example 2.25. Consider $[0, 1]$ and the countable base

$$\{N_r(p) \subseteq [0, 1] \,|\, p, r \in \mathbb{Q}\}.$$

Let x be a point in some open subset S of $[0, 1]$. Accordingly, $N_\varepsilon(x) \subseteq S$ for some $\varepsilon > 0$. Furthermore, $\varepsilon/2 > n^{-1}$ for some $n \in \mathbb{N}$. By the density of the rationals in the reals, there exists some rational $|p - x| < n^{-1}$. Consequently,

$$x \in N_{n^{-1}}(p) \subseteq N_\varepsilon(x) \subseteq S.$$

Exercise 2.26. Prove that every compact metric space K has a countable base, and that K is therefore separable.

Proof. For each n, let $\{N_{n^{-1}}(x_{i,n}) \,|\, 1 \leq i \leq m_n\}$ be a finite subcover of $\{N_{n^{-1}}(x) \,|\, x \in K\}$. Fix $\varepsilon > 0$ and $x \in K$. So, there exists n and i, for which $\varepsilon/2 > n^{-1}$ and $x \in N_{n^{-1}}(x_{i,n})$. Hence $N_{n^{-1}}(x_{i,n}) \subseteq N_\varepsilon(x)$. As such,

$$\bigcup_n \{N_{n^{-1}}(x_{i,n}) \,|\, 1 \leq i \leq m_n\}$$

is a countable base. Furthermore, x is a limit point of the countable dense subset

$$\bigcup_n \{x_{i,n} \,|\, 1 \leq i \leq m_n\},$$

if it is not some $x_{i,n}$.

Exercise 2.27. Define a point p in a metric space X to be a *condensation point* of a set $E \subseteq X$ if every neighbourhood of p contains uncountably many points of E. Suppose $E \subseteq \mathbb{R}^k$, E is uncountable, and let P be the set of all the condensation points of E. Prove that P is perfect and that at most countably many points are not in P. In other words, show that $P^\complement \cap E$ is at most countable.

Proof. We consider E as our metric space for this proof. Clearly, P is closed. So, let x be a limit point of p and $\varepsilon > 0$. Then, there exists $p \in P$ with $|x - p| < \varepsilon/2$. For each of the uncountably many $|p - q| < \varepsilon/2$, we have

$$|x - q| \leq |x - p| + |p - q| < \varepsilon.$$

As such, P is perfect.

Chapter 3

Numerical Sequences and Series

Exercise 3.1. Prove that the convergence of $\{s_n\}$ implies the convergence of $\{|s_n|\}$. Is the converse true?

Proof. Let $\{s_n\}$ be a sequence in a metric space X converging to x, and y any point of X. For $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $d(s_n, x) < \varepsilon$. So by the triangular inequality,

$$|d(s_n, y) - d(x, y)| \leq d(s_n, x) < \varepsilon.$$

Hence, $\{d(s_n, y)\}$ converges to $d(x, y)$.
The converse cannot hold. A counterexample is the alternating series.

Exercise 3.2. Calculate $\lim_{n\to\infty}(\sqrt{n^2+n} - n)$.

Proof. Let $\varepsilon > 0$ and $c := 1/2 - \varepsilon$. Notice that $n^2 + n + 1/4 > n^2 + n$ implies $1/2 - \left(\sqrt{n^2+n} - n\right) > 0$. Moreover, there is some $N > \frac{c^2}{1-2c}$. So, for $n \geq N$,

$$\begin{aligned} \frac{c^2}{1-2c} &< n, \\ n^2 + 2cn + c^2 &< n^2 + n, \\ n + c &< \sqrt{n^2+n}, \\ \left|\frac{1}{2} - \left(\sqrt{n^2+n} - n\right)\right| &< \varepsilon. \end{aligned}$$

In other words, $\lim_{n\to\infty}(\sqrt{n^2+n} - n) = 1/2$.

Exercise 3.3. If $s_1 = \sqrt{2}$, and

$$s_{n+1} = \sqrt{2 + \sqrt{s_n}} \qquad (n = 1, 2, 3, \dots),$$

prove that $\{s_n\}$ converges, and that $s_n < 2$ for $n = 1, 2, 3, \dots$

Proof. Notice that $\sqrt{2+\sqrt{2}} < \sqrt{4} = 2$. So, assume $s_n \leq 2$ for $n = k$, and consider $n = k+1$. Thus,

$$s_{n+1} \leq \sqrt{2+\sqrt{2}} < 2.$$

That is, s_n is always less than 2.

Furthermore, since $\{s_n\}$ is nonnegative, it is monotonically increasing. By the Monotone Sequence Theorem (3.14), $\{s_n\}$ converges.

Chapter 4

Bibliography

1. Cover page is modified from https://tex.stackexchange.com/a/85989.
2. Diagram on cover page: https://tex.stackexchange.com/a/525667.